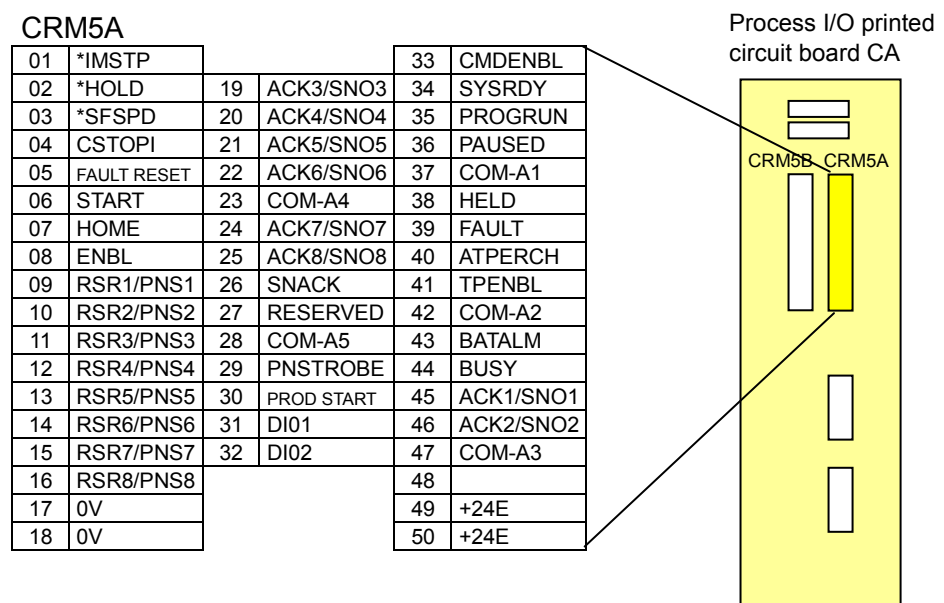


15.2 PERIPHERAL DEVICE I/O

The peripheral device I/O (UI/UO) is a special signal with a defined purpose. These signals are used to externally control a robot by accessing the Remote controller, a peripheral device, etc. via the process I/O printed circuit board (or I/O unit).

The assignment of signals to pin numbers depends on the I/O assignment.

The following shows an example of pin assignment with the process I/O printed circuit board CA.



15.2.1 Overview of Special Input Signals

*** IMSTP UI[1]**
(immediate stop)

Immediate stop signal: A normally closed signal.

The immediate stop signal specifies an emergency stop via the software.

When this signal is off (the contact is open), the following processing is performed.

- An alarm to shut down the servo power supply is generated.
- The operation of the robot is stopped immediately and program execution is suspended.

Although this signal has the same function as an emergency stop, it is not recommended to use this signal to ensure safety as the control is performed by software. To directly block the motor power line for safety purposes, do so by linking it with an emergency stop. Use it along with the EMGIN1 and EMGIN2 signals on the operator's panel printed circuit board. For details on these signals, refer to the maintenance manual.

*** HOLD UI[2]**
(hold)

Hold signal: A normally closed signal.

The hold signal specifies a hold from an external device.

When this signal is off (the contact is open), the following processing is performed.

- The running operation is decelerated and stopped, and program execution is suspended.
- If "Brake on hold" is enabled in the General Item settings, the servo power supply is shut off by generating an alarm after the robot stops.

While this signal contact is open, the "HELD" signal is output.

*** SFSPD UI[3]**
(safety speed)

Safe speed signal: A normally closed signal.

The safety speed signal suspends the robot when the safety fence is released. This signal is normally connected to the safety plug of the safety fence.

When this signal is off (the contact is open), the following processing is performed.

- The running operation is decelerated and stopped, and program execution is suspended. The program being executed will be set to on hold. In addition, the override value drops to the set value and is limited so that it does not exceed that value.
- When *SFSPD input is off, if a program is launched from the teach pendant, the feedrate override will be lowered to the set value. When jog feed is executed, it will also be lowered to the set value. You cannot increase the feedrate override above the specified value.

The *SFSPD signal controls deceleration and stop by software. To stop the robot in an emergency use this signal along with the FENCE1 and FENCE2 signals on the operator's panel printed circuit board. For details, refer to the maintenance manual.

● **Related system variables**

\$SCR.\$FENCEOVRD: When the *SFSPD contact is open, the feedrate override is limited to a value below this value.

\$SCR.\$SFJOGOVRD: When the *SFSPD contact is open, if jog feed is performed, the feedrate override is limited to a value below this value.

\$SCR.\$SFRUNOVRD: When the *SFSPD contact is open, if the program is run, the feedrate override is limited to a value below this value.

\$SCR.\$RECOV_OVRD: When the *SFSPD contact is closed, the feedrate override is restored to the original value. However, the following conditions must be met.

- \$SCR.\$RECOV_OVRD is true.
- In a remote state.
- The feedrate override has not been changed while the SFSPD contact was open.

CSTOPI UI[4]

(cycle stop input)

Cycle stop signal

This signal clears the program that is in a wait state due to RSR.

The currently running program will be as follows according to the "CSTOPI for ABORT" setting on the "6 SYSTEM.Config" system configuration screen.

When "DISABLE":

- The currently running program continues to execute. The program ends after completion. (Standard setting)

When "ENABLE":

- Immediately halts the program currently being executed.

FAULT_RESET UI[5]

(fault reset)

Alarm reset signal

If this signal is input when the cause of the alarm has been removed, the alarm state is reset. When the servo power is shut off, it is started. This signal functions during standard startup. At the same time, the alarm display appearing at the top section of the teach pendant disappears.

The input of this signal only resets the alarm state. Even if this signal is turned on when the robot is in operation, the robot will continue to operate.

START UI[6]

(start)

External startup signal (valid in a remote state)

This signal functions at its falling edge when turned off after being turned on. When this signal is received, the following processing is performed.

When "FALSE":

- The program selected using the teach pendant is executed from the line to which the cursor is positioned. A temporarily stopped program is also continued. (Default)

When "TRUE":

- A temporarily stopped program is continued. When the program is not temporarily stopped, it cannot be started.

Note) To start a program from a peripheral device, RSR input or PROD_START input is used although it also depends on the automatic operation system. To resume a halted program, start input is used.

HOME UI[7]	This signal is currently not used. It is only available for macro functions.
ENBL UI[8] (enable)	Operation enable signal: A normally closed signal. This signal enables robot operation to set it to the operation enabled state. When this signal is off (the contact is open), the robot's operation including jog feed, DO output, and program playback are prohibited. Also, the program playback will be set to on hold.
RSR1 to RSR8 UI[9] to UI[16] (robot service request)	Robot start request signals (valid in a remote state) These are automatic operation start signals. When one of these signals is received, a set RSR program corresponding to that signal is selected and the program always starts from the first line. When other programs are running or on hold, the selected program is added to the wait queue and started after the running program is completed. In addition, depending on the setting, when each RSR signal is input, RSR receiving acknowledgment signals (ACK1 to ACK8) is output.
PNS1-8 UI[9] to UI[16] (program number select)	Program number selection signals (valid in a remote state) When using program selection, you can select the program to be executed by combining these signals.
PNSTROBE UI[17] (program number select strobe)	PN strobe signal (valid in a remote state) This signal is used in external program selection. The status of PNS1 to PNS8 when the PNSTROBE signal is input is read as a binary code to select the program to be executed. When other programs are being executed or on hold, this signal will be ignored. Also, when the remote condition has been satisfied and PNSTROBE is turned on, program selection on the teach pendant is prohibited.
PROD_START UI[18] (production start)	Automatic operation start signal (valid in a remote state) When used in combination with PNS, this signal executes the program selected by PNS from the first line. When this signal is not used in combination with PNS, the program selected by the teach pendant is executed from the first line. When other programs are being executed or on hold, this signal will be ignored. This signal functions during shutdown after it has been turned on once.

15.2.2 Overview of Special Output Signals

CMDENBL UO[1]

(command enable)

Input receiving enable signal

This signal is output when the following conditions are met. This signal indicates that a program can be started including a motion (group) from the Remote controller.

- Remote condition has been satisfied.
- Operation-enabled condition has been satisfied.
- In a continuous operation state (rather than step operation)

• **Remote condition:**

Starting a program with the peripheral device I/O input requires the robot to be in a remote state. A remote state is valid when the following conditions are met.

- The teach pendant enable switch is turned off.
- The Remote/Local setting on the system settings screen is set to Remote. Or, the Remote/Local switch on the operator's panel is set to remote.
- System variable \$RMT_MASTER is set to 0 (peripheral device).
- The *SFSPD signal is turned on (the contact is closed).

• **Operation-enabled condition:** The startup of a program that contains a motion (group) requires that the following operation-enabled conditions be met.

- The ENBL signal is turned on (the contact is closed).
- The servo power supply is started (not in an alarm state).

SYSDRY UO[2]

(system ready)

System preparation completion signal

This signal is output when the servo power supply is started (the motor is energized). In an operation-enabled state, executing jog feed and starting a program including the operation (group) are possible. In an operation-enabled state, the following operation-enabled conditions are satisfied.

- The ENBL input of the peripheral device I/O is turned on.
- The servo power supply is started (not in an alarm state).

PROGRUN UO[3]

(program run)

Program run signal

This signal is input during program playback (running). It will not be output when the program is on hold.

PAUSED UO[4]

(paused)

Halt signal

This signal is output when the program is on hold and waiting to be resumed.

HELD UO[5]

Held signal

(held)	This signal is output when the HOLD button is being pressed and the HOLD signal is input (the contact is open). When the HOLD button is released, this signal will not be output.
FAULT UO[6] (fault)	Alarm signal This signal is output when the Robot controller is in an alarm state (when an alarm that prevents program execution is detected). However, it will not be output in a warning (WARN alarm) state. The alarm is reset by the input of FAULT_RESET.
ATPERCH UO[7] (atperch)	Reference position signal This signal (atperch) is output when the robot is at the first reference point defined in advance. Up to 10 reference points can be set, and the general purpose signal is assigned for additional reference points.
TPENBL UO[8] (teach pendant enable)	Teach pendant enable signal This signal is output when the teach pendant enable switch is turned on.
BATALM UO[9] (battery alarm)	Battery abnormal signal The battery abnormal signal (battery alarm) is an alarm for when the voltage of the controller backup battery is low. An alarm for when the voltage of the pulse coder backup battery for the Robot mechanical unit is low can also be set to display.
BUSY UO[10] (busy)	Busy signal This signal is output when the program is being executed or the processing on the teach pendant, etc. is in progress. (The signal is output when the "BUSY" lamp on the teach pendant is on.) It will not be output when the program is on hold.
ACK1-8 UO[11] to UO[18] (acknowledge)	RSR receiving acknowledgment signals These signals are used in combination when the RSR function is enabled. When RSR input is received, the corresponding signal from RSR1 to RSR8 is output as a pulse signal as acknowledgment. The pulse width can be specified in the settings.
SNO1-8 UO[11] to UO[18] (selected number outputs)	Selected program number signals These signals are used in combination when the PNS function is enabled. The currently selected program number (the signal corresponding to PNS1 to PNS8) is always output as a binary code (SNO1 to SNO8) for acknowledgment. SNO1 to SNO8 are rewritten when a new program is selected. These signals are used to check the program number with the PLC (programmable logic controller) when the program external selection function (option) is enabled.
SNACK UO[19]	PNS receiving acknowledgment signals

(selected number acknowledge)

These signals are used in combination when the PNS function is enabled. When PNS input is received, a pulse signal is output for acknowledgment. The pulse width can be specified in the settings.

15.3 END EFFECTOR SIGNALS

The interface between a controller and an end effector is connected using the connector provided for the mechanical unit. It is a digital signal called the robot I/O. The robot I/O consists of up to eight input and output points of general purpose signals. For these signals, signal numbers cannot be redefined.

Depending on the robot model, the number of general purpose input/output points for the robot I/O varies. For details, refer to the mechanical unit manual for each robot.

In addition to RI/RO, *HBK and *PPABN are provided.

Example of an end effector interface (RI/RO):
(in the case of R-2000/B)

	1	2	3	4	
	RO1	RO2	RO3	RO4	
5	6	7	8	9	
RO5	RO6	*HBK	0V	RI1	
10	11	12	13	14	15
RI2	RI3	RI4	RI8	*PPABN	RI5
16	17	18	19	20	
RI6	+24E	+24E	+24E	+24E	
	21	22	23	24	
	RO7	RO8	0V	RI7	

* HBK

(Hand Broken)

Hand broken signal: A normally closed signal.

This signal detects a broken tool (hand, etc.) of the robot by accessing it. The *HBK signal is turned on (the contact is closed) in a normal state. When the signal is on (the contact is open), the robot issues an alarm and comes to an emergency stop.

If a breakage is detected, the following alarm is displayed.

"SRVO—006 Hand Broken"

To use this signal, connect the limit switch and proximity sensor for detecting a hand broken condition to this signal line.